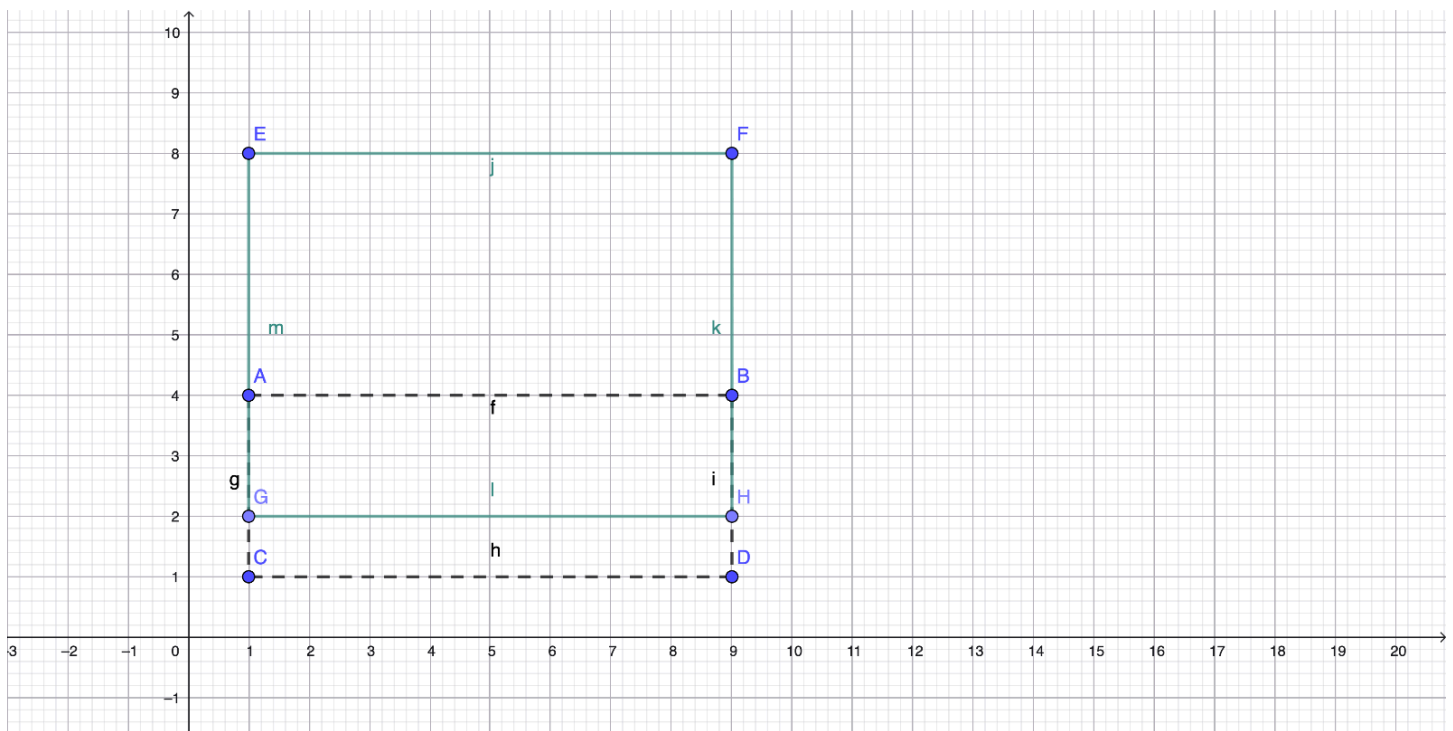


Stretch



I did a stretch along the y-axis of the rectangle ABCD, by a factor of 2

The transformation matrix is:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

With it applied to each coordinate A,B,C and D to produce E,F,G and H:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1(1) + 0(4) \\ 0(1) + 2(4) \end{pmatrix} = \begin{pmatrix} 1 \\ 8 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 9 \\ 4 \end{pmatrix} = \begin{pmatrix} 1(9) + 0(4) \\ 0(9) + 2(4) \end{pmatrix} = \begin{pmatrix} 9 \\ 8 \end{pmatrix}$$

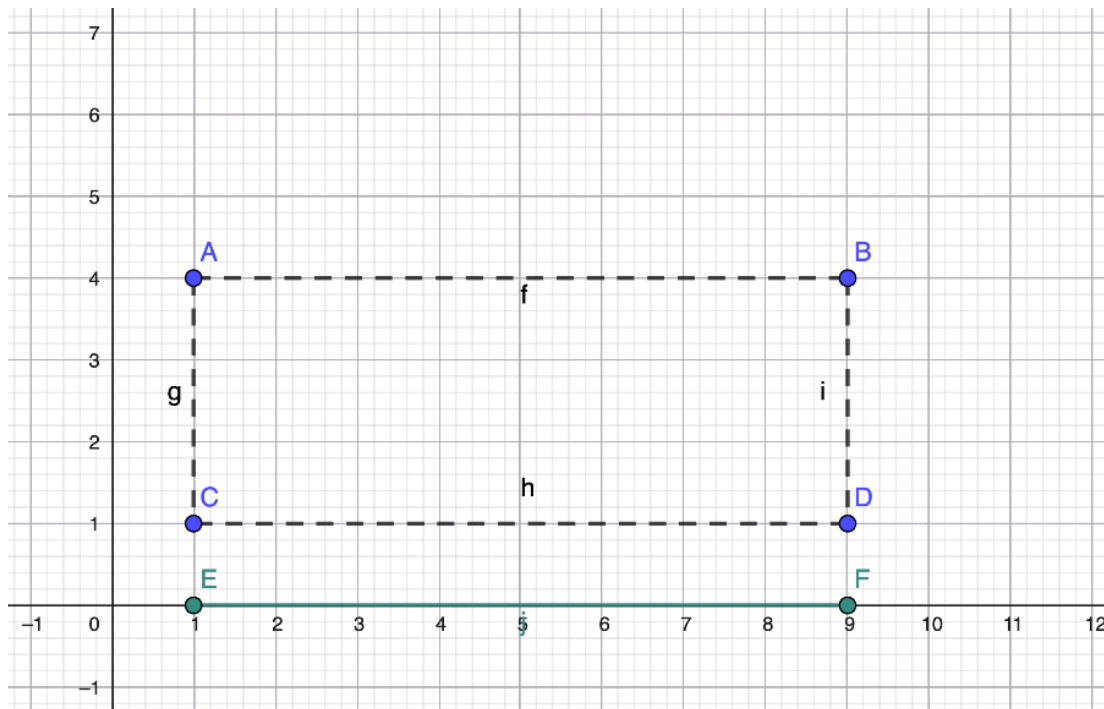
$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1(1) + 0(1) \\ 0(1) + 2(1) \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 9 \\ 1 \end{pmatrix} = \begin{pmatrix} 1(9) + 0(1) \\ 0(9) + 2(1) \end{pmatrix} = \begin{pmatrix} 9 \\ 2 \end{pmatrix}$$

We can also see from the determinant of the transformation $\det(ABCD) = (1)(2) - (0)(0) = 2$, so every region's area is multiplied by 2, as we can see from the rectangle.

Orthogonal Projection

A projection of rectangle ABCD on the x-axis.



After applying the projection transformation: A and C both land on E, while B and D both land on F, as can be seen below:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1(1) + 0(4) \\ 0(1) + 0(4) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 9 \\ 4 \end{pmatrix} = \begin{pmatrix} 1(9) + 0(4) \\ 0(9) + 0(4) \end{pmatrix} = \begin{pmatrix} 9 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1(1) + 0(1) \\ 0(1) + 0(1) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 9 \\ 1 \end{pmatrix} = \begin{pmatrix} 1(9) + 0(1) \\ 0(9) + 0(1) \end{pmatrix} = \begin{pmatrix} 9 \\ 0 \end{pmatrix}$$

This projection is orthogonal because each point moves perpendicular to the x-axis, so e.g. $A(1,4)$ keeps its x-coordinate and just loses its height to become $(1,0)$.